NATIONAL CHEMISTRY CONTEST 2023

"Every aspect of the world today – even politics and international relations – is affected by chemistry."

Linus Pauling

Name(s):______________________________ Last name:______________________________

School:____

Municipality:______________________________ Province:______________________________

CODE:__________________________________ Exam: **Day 2. TWELFTH GRADE**

Instructions

- Write clearly all your data in the appropriate place (names, last names, school, municipality, province). Do not write anything in the field CODE.
- Verify that your questions booklet is complete. This is the Day 2 problem set, with an individual exam for 11th grade. The exam has 3 problems (problem 12, 5 items; problem 15, 5 items; problem 16, 18 items) and a total of 8 pages.
- All data, constants, conversions and basic equations needed are included in the exam. You are allowed to use a scientific calculator.
- At the beginning of each problem its scoring (from a total of 100 points), and each item scoring (in marks) is indicated.
- Number the lower part of the exam, each page and indicate the number of total answer pages, for example, "1 of 7", "2 of 7" and so on.
- You are only given four and a half hours, control timing carefully: pay attention that the exam has a large number of questions. Do not waste time on unnecessary answers!

Success!

Ministerio de Educación

República de Cuba

FACULTAD DE QUÍMICA
UNIVERSIDAD DE LA HABANA

CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6{,}0221 \cdot 10^{23}\ mol^{-1}$	Ideal gas law	$p \cdot V = n \cdot R \cdot T$
Gasses' constant	$R = 8{,}3145\ J \cdot K^{-1} \cdot mol^{-1}$	First Principle of Thermodynamics	$Q = \Delta U + W$
Zeroth in the Celsius's scale	$273{,}15\ K$	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485\ C \cdot mol^{-1}$	Nernst' equation	$E = E° - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{red}}{c_{ox}}$
Water ionic product at 25 °C	$K_W = 1{,}00 \cdot 10^{-14}$		$\Delta G° = -R \cdot T \cdot \ln K° = -z \cdot F \cdot \Delta E°$
Standard conditions	$p° = 100\ kPa = 1\ bar$ $c° = 1\ mol \cdot L^{-1}$	Van't Hoff i factor	$i = 1 + \alpha \cdot (\upsilon - 1)$
Arrhenius' equation	$k = A \cdot \exp\left(-\frac{E_A}{R \cdot T}\right)$	Cryoscopy	$\Delta T_c = i \cdot K_c \cdot b$

$1\ J = 1\ C \cdot 1\ V = 1\ kPa \cdot 1\ L = 1\ W \cdot 1\ s$ $\quad$ $1\ C = 1\ A \cdot 1\ s$ $\quad$ $1\ dm^3 = 1\ L = 1000\ mL$

1 atm = 760 Torr = 760 mmHg = 101,325 kPa $\quad$ $1\ eV = 96{,}485\ kJ \cdot mol^{-1}$ $\quad$ $M \equiv mol/L$

PERIODIC TABLE OF THE ELEMENTS

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 H 1.008																	2 He 4.003
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.97	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.95	43 Tc -	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57-71	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po -	85 At -	86 Rn -
87 Fr -	88 Ra -	89-103	104 Rf -	105 Db -	106 Sg -	107 Bh -	108 Hs -	109 Mt -	110 Ds -	111 Rg -	112 Cn -	113 Nh -	114 Fl -	115 Mc -	116 Lv -	117 Ts -	118 Og -

57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm -	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
89 Ac -	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np -	94 Pu -	95 Am -	96 Cm -	97 Bk -	98 Cf -	99 Es -	100 Fm -	101 Md -	102 No -	103 Lr -

Problem 12. Chemical equilibrium (only 11th and 12th grades)

12.1	12.2	12.3	12.4	12.5	Total	Total
9	9	8	7	7	40	10%

12.1. An 18,92% by mass aqueous solution of a strong and soluble metal chloride was prepared. The change in the melting point of the solution from that of pure water was about 13,68 °C. Identify the metal chloride if it is known that K_{cryos} (H_2O) = 1,86 °C·kg·mol^{-1}.

12.2. Three gases A, B, and C are in equilibrium according to $A(g) \rightleftharpoons B(g) + C(g)$. A certain amount of A is introduced into an empty container. If the K_p at 298 K is 4,00 and the total pressure of the gases in equilibrium is 2,00 bar, calculate the ratio between the amounts of substance of B and of A in balance.

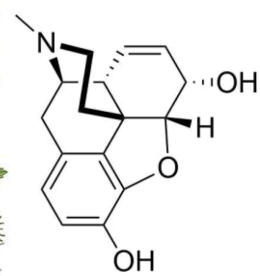
12.3. Calculate the pH of the solution resulting from mixing 25 mL of NH_3 0,01 mol·L^{-1} with 15 mL of HCl 0,01 mol·L^{-1}. K_b (NH_3) = 1,8·10^{-5}

12.4. If the solubility of silver oxalate at 25 °C is 1,11·10^{-4} mol·L^{-1}, calculate the solubility product of the salt. Neglect any hydrolysis of the oxalate ion.

12.5. Calculate the solubility product constant of Hg_2Cl_2 in water at 25°C if the reduction redox potentials are E° ($[Hg_2]^{2+}$/Hg(l)) = 0,789 V and E° (Hg_2Cl_2(s)/Hg(l), Cl^-) = 0,268 V.

Problem 15. (-)-Morphine (12th grade)

15.1	15.2	15.3	15.4	15.5	Total	Total
20	4	8	4	4	40	10%



Morphine is the main alkaloid-like component of opium, a complex mixture of substances extracted from the capsules of the opium poppy (*Papaver somniferum*). This compound has great relevance in medicine and its structure has presented a challenge for the art and science of organic synthesis. In 2002, Rheingold's group proposed a synthetic methodology for said alkaloid. (J. Am. Chem. Soc. 2002, 124(42), 12416-12417). A fragment of the synthetic sequence used is shown on the next scheme.

Answer the following questions.

15.1. Represent the structures of compounds **C**, **D**, **E**, **F1**, **F2**, **G**, **H**, **I**, **J**, **K** and **L**.

15.2. Propose which (non)organic substance(s) could be used as reagents **A**, **B** and **M**.

15.3. Propose possible mechanisms for

15.3.a) the first reaction step

15.3.b) the transformation of alcohol into azide, if it is known that triphenylphosphine oxide is formed as a side product and one of the intermediates formed has two N-P bonds in its structure.

15.4. About the sequence of reactions answer:

15.4.a) What kind of isomers are **F1** and **F2**?

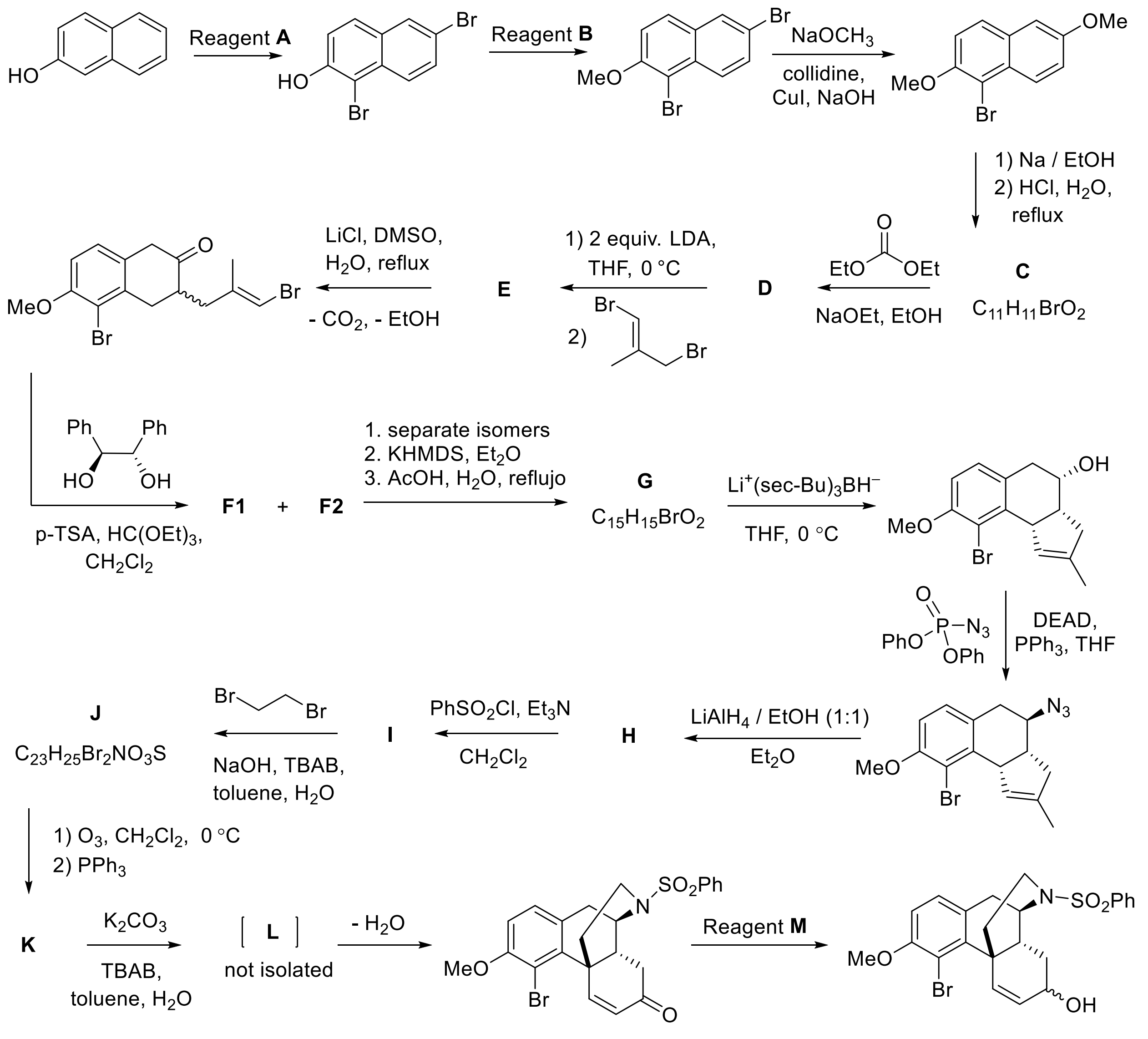
15.4.b) Why is the reaction with lithium tris-*sec*-butylborohydride stereoselective?

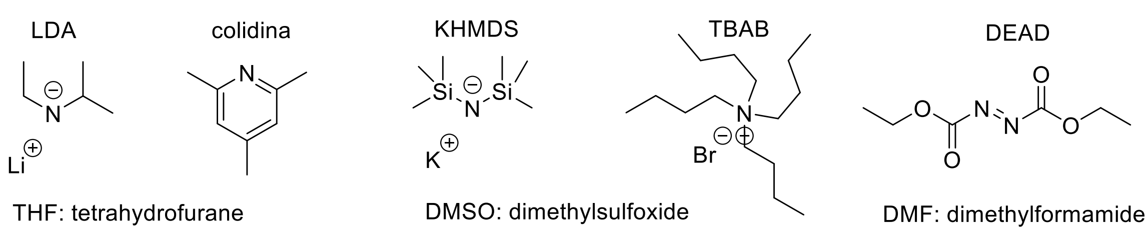
15.4.c) For what purpose is benzenesulfonyl chloride used?

15.4.d) What is the function of copper(I) iodide?

15.5. Propose a method to obtain (1S,2S)-1,2-diphenylethane-1,2-diol (as a racemic mixture, together with its enantiomer) from the appropriate 1,2-dephenylethene isomer.

15.5.a) Represent the Newman projection of the most stable conformation of this diol, knowing that there is an intramolecular hydrogen bond that stabilizes it.



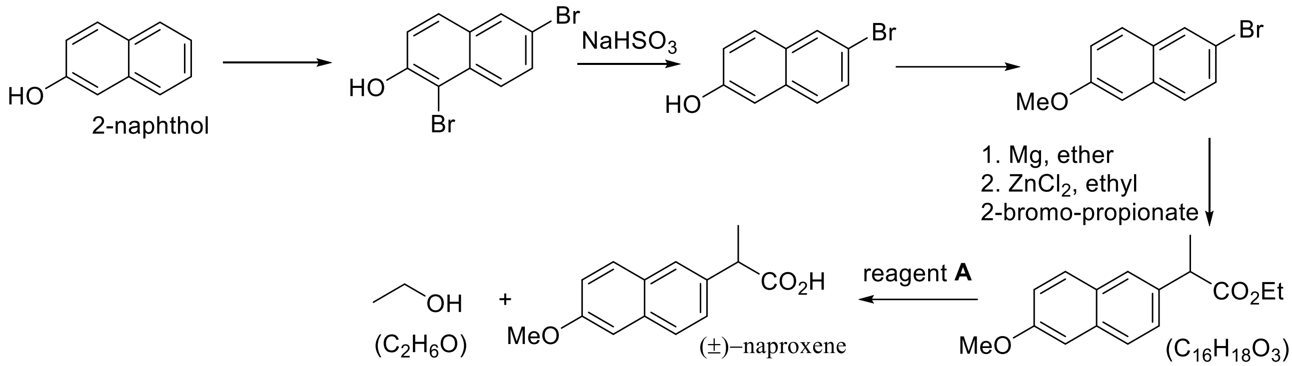


THF: tetrahydrofurane DMSO: dimethylsulfoxide DMF: dimethylformamide

Problem 16. Aromatic compounds (12th grade)

16.1	16.2	16.3	16.4	16.5	16.6	16.7	16.8	16.9	16.10	16.11	16.12	16.13	16.14	16.15	16.16	16.17	16.18	Total	Total
10	6	11	20	2	8	2	8	2	5	4	3	6	6	6	5	10	6	120	30%

A large part of the known organic compounds have at least one aromatic ring in their structure, so the study of their production methods and chemical properties is of great importance. Naproxen is an orally administered non-steroidal anti-inflammatory drug used to treat pain, rheumatoid arthritis, gout and fever and can be obtained according to the sequence.



16.1. About the above sequence, please answer.

16.1.a) Naproxen is obtained in the form of a racemate, in which reaction did the racemization occur?

16.1.b) Illustrate the structure of the S stereoisomer of naproxen, which is the active compound.

16.1.c) How could you separate the S isomer from the R isomer?

16.1.d) Propose a qualitative test to identify the presence of the acid group in the laboratory, if you have a sample of naproxen and additional necessary inorganic reagents.

16.1.e) Propose possible reagents **A** that could be used to carry out the last reaction.

The atom economy of a chemical process is a parameter used to quantify how environmentally friendly or "green" a synthesis is. It is defined as the ratio between the masses of the main product and the sum of the masses of all the reactants, assuming that the reaction is quantitative.

$$\text{atom economy} = \frac{\text{m(main product)}}{\sum \text{m(reactant)}} \cdot 100\%$$

16.2. Catalysts and solvents used in chemical reactions are not included in the atomic economy calculation.

16.2.a) Why are these substances not included in the calculation?

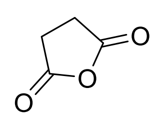
16.2.b) Calculate the atom economy of the last reaction step.

16.3. Describe possible synthetic routes to obtain the 2-naphthol used in the synthesis of naproxen from

16.3.a) Naphthalene from processed coal tar.

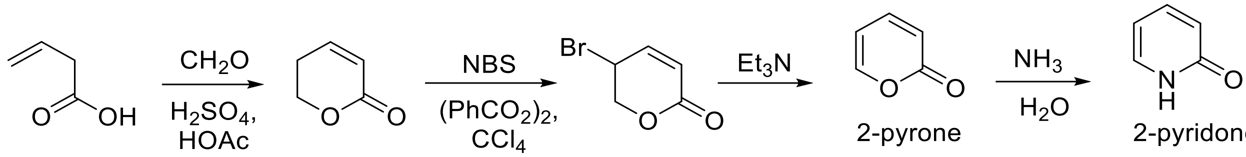
16.3.b) Methoxybenzene and succinic anhydride.

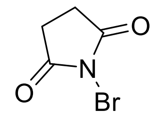
16.3.c) Which method do you consider to be more efficient?



anhídrido succínico

2-Pyrone is a very useful synthetic intermediate which can be obtained according to the following sequence and is converted to 2-pyridone when treated with concentrated aqueous ammonia solution.





16.4. About 2-pyrone and 2-pyridone, please answer.

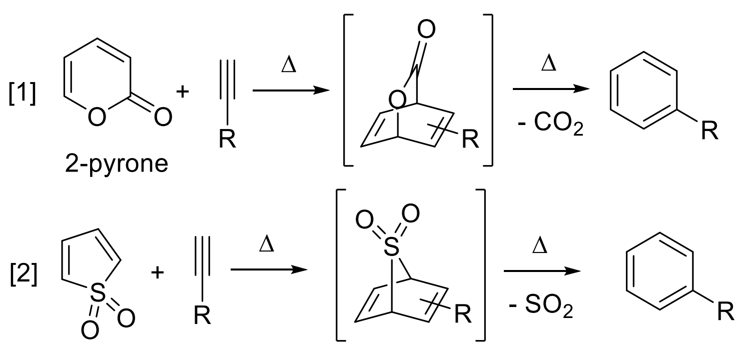
16.4.a) Propose a mechanism for the last three reaction steps.

16.4.b) Represent the structures of the aromatic tautomers of both compounds. Why is the non-aromatic structure the most favored in both cases?

16.4.c) 2-Pyrone can undergo electrophilic aromatic substitution reactions. What ring positions are most activated for this type of reaction? Justify your answer by representing the most relevant resonant structure of the corresponding sigma complex.

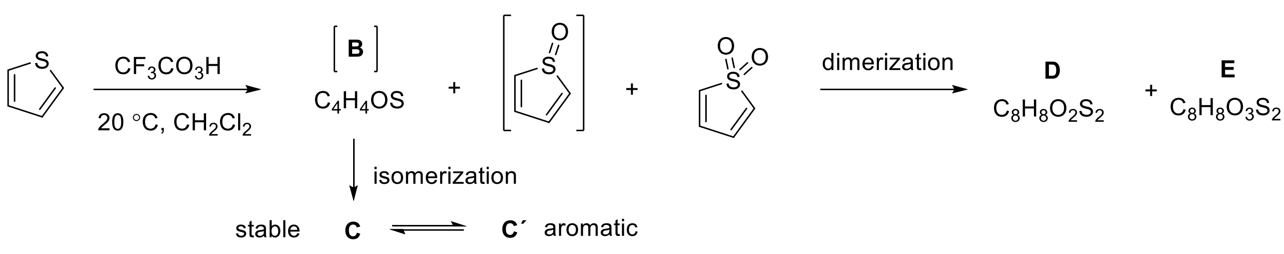
16.4.d) When 2-pyridone dissolves in nonpolar solvents it forms a dimer in which there are two hydrogen bonds. represent its structure.

2-Pyrone, thiophene, as well as the sulfoxide and sulfone derivatives thereof are dienes and give rise to the Diels-Alder [4+2] cycloaddition reaction when reacted with suitable dienophiles. Synthetic sequences that include a cycloaddition step followed by a back-cycloaddition to remove a gas (which is favored entropically) have become standard methodology for obtaining and modifying aromatic systems, as illustrated in the following examples.



16.5. The tendency to undergo the Diels-Alder reaction is increased in the sequence thiophene < sulfoxide < sulfone. How do you explain this variation in reactivity?

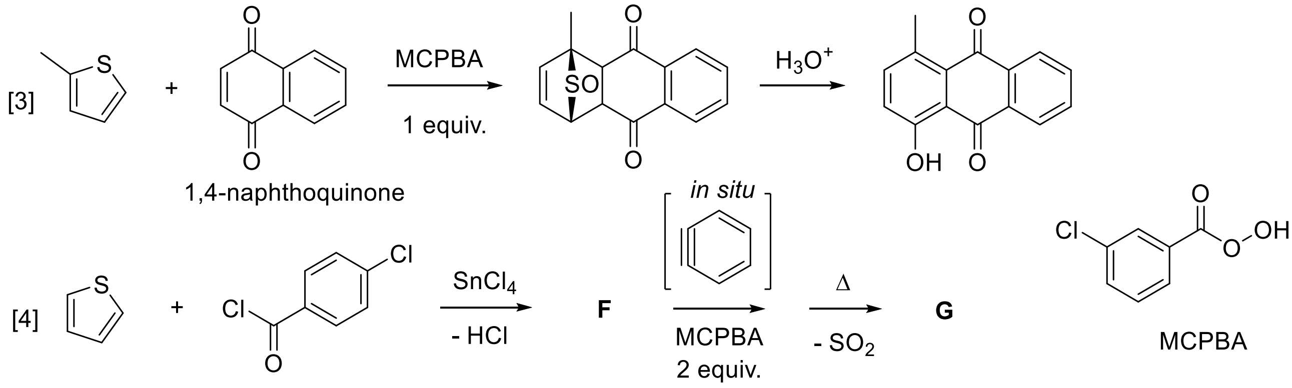
Oxidation of thiophene to the corresponding sulfoxide or sulfone can be accomplished by adding one or two equivalents, respectively, of hydrogen peroxide, oxone, or an organic sulfate peroxide. However, it is a complex process in which a series of side reactions occur, forming unwanted oxidation, isomerization and cycloaddition products.



16.6. Represent the structures of **B**, **C**, **D**, **E** and the aromatic tautomer **C´**.

16.7. Draw the formula for the peroxymonosulfate anion present in the triple salt Oxone $2KHSO_5 \cdot KHSO_4 \cdot K_2SO_4$ if it is known to form an intramolecular hydrogen bond.

To minimize these undesirable processes, thiophene is usually oxidized *in situ*, in the presence of the dienophile with which it is going to react, as occurs in the following reaction sequences.



16.8. Propose a mechanism for the hydrolysis of the tricycle forming a 1,4-naphthoquinone derivative.

16.9. One of the reactants used is benzine. How can you get this intermediate *in situ*?

16.10. Propose a method for preparing 4-chlorobenzoyl chloride from toluene.

16.11. Represent the structures of **F** and **G**.

16.12. 1,4-naphthoquinone is a very toxic compound for the organism because it reacts with the thiol groups present in the glutathione tripeptide and in proteins, causing liver damage. Write down the reaction that occurs and say its name.

In one of the above routes, 1,4-naphthoquinone, an oxidation product of naphthalene, is used as a starting material. Naphthalene can also be oxidized to another **H** product with the formula $C_8H_4O_3$ through the action of vanadium(V) oxide, in the so-called Gibbs-Wohl reaction. **H** has a symmetrical structure, it reacts with sodium hydroxide and amines. One molecule of **H** adds two molecules of phenol in an acid medium for the same carbon atom, with loss of water. This reaction can also be considered as two consecutive S_EAr processes. The product is the acid-base indicator phenolphthalein.

16.13. Represent the structures of **H** and phenolphthalein.

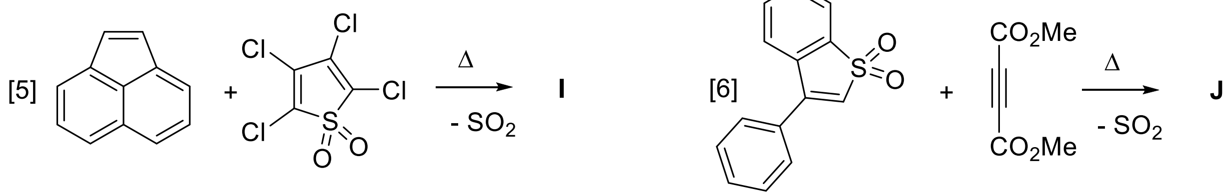
16.13.a) Represent the structure of the anion derived from phenolphthalein if it is known that there is one less ring in its structure, compared to the neutral molecule.

The Diels-Alder reaction can be classified as normal electron demand (NED) when the diene donates electrons to the dienophile, reverse electron demand (IED) when the diene accepts electrons from the dienophile, or intermediate type when there is virtually no charge transfer.

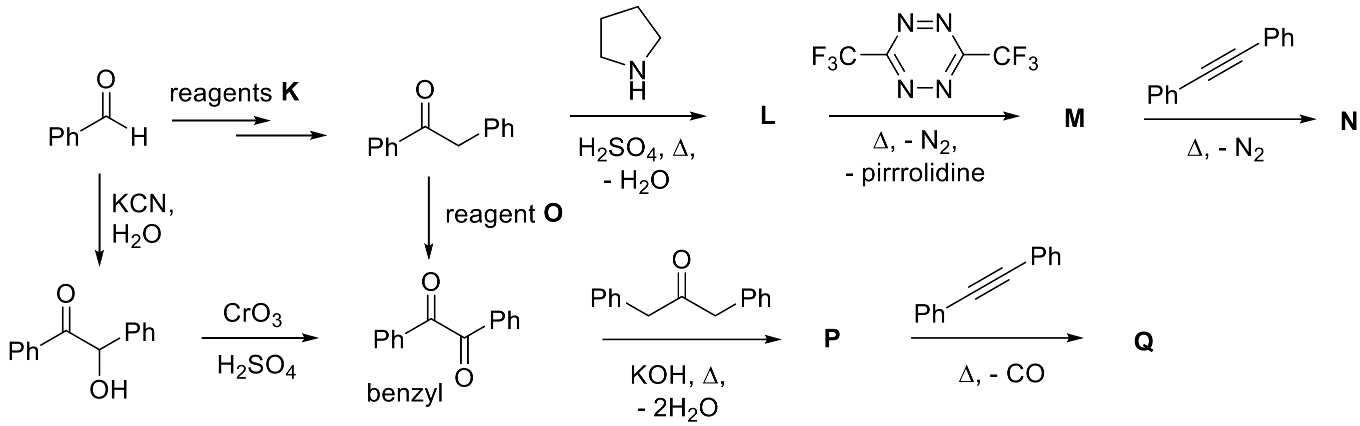
16.14. For the following reactions

16.14.a) Classify them as NED, IED or intermediate type cycloadditions.

16.14.b) Represent the structures of the aromatic products **I** and **J**.



The following synthetic routes allow the formation of aromatic compounds with high symmetry. All use one or more (retro)-Diels-Alder reactions.

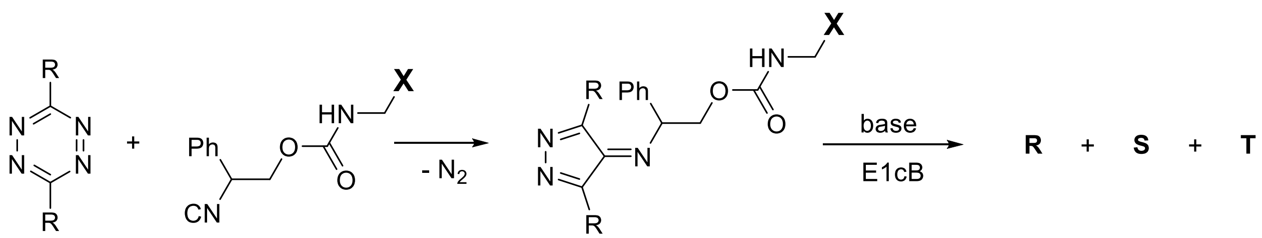


16.15. Propose reagents or a mixture of reagents **K** and **O** to carry out the given transformations.

16.16. Propose reaction mechanisms for the two steps of transformation of benzaldehyde into benzyl diketone.

16.17. Represent the structures of **L**, **M**, **N**, **P** and **Q**.

The 2022 Nobel Prize in Chemistry was awarded to the Dane Morten Meldal and the Americans Carolyn Bertozzi and Barry Sharpless, who already had a previous award, for "the development of Click chemistry and bioorthogonal chemistry", according to the jury of the Royal Swedish Academy of Sciences in its decision. A reaction that recently began to be used for applications of this type is the cycloaddition [4 + 1] between 1,2,4,5-tetrazines and isonitriles. The following scheme illustrates the use of this reaction for the activation of a protein **X** that contains an amino group of a lysine amino acid that is temporarily blocked in the carbamate form. (ACS Omega 2020, 5, 40, 25505-25510).



16.18. Propose structures for the products **R**, **S** and **T**.